

Rafi Mahmud Tumon

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Education

Technical University Berlin

Berlin, Germany

Master of Science in Geodesy and Geoinformation Science

October 2020 – Present

- Specialization: Geoinformation Science; Key Coursework: Photogrammetric Computer Vision, Geoinformatics, Geodatabases and Infrastructures, Space Geodesy, Adjustment Theory, Deep Learning for Geographic Data

Rajshahi University of Engineering and Technology

Rajshahi, Bangladesh

Bachelor of Science in Urban and Regional Planning

March 2014 – November 2018

Work Experience

German Aerospace Center (DLR)

Berlin, Germany

Student Assistant

January 2026 – Present

- Application of machine learning methods with multimodal geodata in urban areas.
- Utilize Python and QGIS for geospatial analysis and model development.

OXG Glasfaser GmbH

Berlin, Germany

Working Student, Strategic Area Development

June 2024 – Present

- Improved operational efficiency by 20% through automated geospatial analysis using QGIS and Python for fiber network deployment.
- Conducted cost-benefit analyses and site assessments to identify optimal deployment areas while navigating geographical constraints and market competition.
- Developed Python-based geocoding tool for German addresses, reducing processing time by 30%.

Department of Civil Engineering, RUET

Rajshahi, Bangladesh

Project Assistant (GIS and Geospatial Planner)

February 2020 – April 2020

- Led an airport master planning project using GIS analysis to improve regional connectivity for underserved communities in northwestern Bangladesh.
- Collaborated with engineering teams and stakeholders to integrate spatial analysis into transportation planning decisions.

Research Experience

Flood Vulnerability Assessment Using Geospatial Analysis

November 2018

Bachelor's Thesis: "Application of Remote Sensing and GIS for Vulnerability Assessment: A Case Study of Bogra, Sirajganj, and Tangail Districts"

- Conducted vulnerability assessment for three flood-prone districts in Bangladesh using satellite imagery to identify at-risk communities.
- Developed geospatial methodologies and produced flood risk maps to support disaster planning.

Personal Projects

Forest Loss Analysis in Colombia's Indigenous Lands

github.com/rafimt/Deforestation-Analysis

- Analyzed 10 years of satellite images using Google Earth Engine and Python to track deforestation and its effects on vulnerable communities, revealing a 15% loss in tree cover.

Wildfire Data Analysis

github.com/rafimt/Wildfire_Analysis_2022

- Developed a Python tool to analyze wildfire patterns in 8 countries (analyzing 1,234 fires total) to help with disaster planning.

Berlin's Environmental Dynamics Analysis

github.com/rafimt/Project-Geoinformatics-2024

- Assessed Urban Heat Island effects using satellite data, finding that built-up areas experienced the highest temperatures compared to other land types.

Skills and Tools

- **Core Technical:** GIS, Remote Sensing, Spatial Analysis, Geodatabase Maintenance, Spatial Data Management, Field Research & Data Collection
- **QGIS:** Data Management, Coordinate Reference Systems, Map Creation and Cartography, Spatial Joins, Plugin Usage, Database Connectivity, Processing Modeler, Python Scripting and Workflow Automation
- **Programming:** Python (Pandas, Geopandas, Rasterio, OSMnx, Folium, scikit-learn, Seaborn, PyTorch, Matplotlib), SQL, JavaScript
- **Tools & Software:** Google Earth Engine, Git, Jupyter Notebook, Excel, PostgreSQL/PostGIS, MATLAB
- **Web GIS:** Django, HTML, CSS, Leaflet, Bootstrap
- **Machine Learning:** Supervised/Unsupervised Learning, Neural Networks, Model Evaluation, Classification, Object Detection, CNN, Vision Transformer
- **Geospatial Expertise:** Engineering Survey, Least Squares Adjustment, Cartographic Principles, GNSS Processing, Environmental Analysis, Vulnerability Assessment

Awards and Recognition

- **Planning and Design of Compact Township, 2nd Runner Up**
Inter-University Undergraduate Planning Students Competition, Bangladesh University of Engineering and Technology (BUET)
- **Publication**
“Identification of Determinant Factors Influencing Modal Choice Behavior and Satisfaction Level of Commuters: A Case of Rajshahi City”
Trends in Civil Engineering and its Architecture, Lupine Publishers
DOI: 10.32474/TCEIA.2019.03.000171

Languages

English (Professional working proficiency), **German** (Elementary proficiency - A2)